# Yueci Deng

Shenzhen, China

Systems, Simulation & Embodied Intelligence

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## EDUCATION

**The Chinese University of Hong Kong, Shenzhen** 2025 - Present
Ph.D. in Computer Science, School of Data Science. Supervisor: Prof. Kui Jia.

**Nanyang Technological University, Singapore** Jul 2018 - Jun 2019
M.Sc. in Signal Processing, Electrical and Electronic Engineering.

**UESTC / University of Glasgow** Sep 2014 - Jun 2018
Joint B.Eng. programme: Electronic Information Engineering (UESTC); Electronics and Electrical Engineering (Glasgow), First Class Honours.

## INDUSTRY EXPERIENCE

**DexForce Technology Co., Ltd.** Aug 2022 - Aug 2025
Head of Core Engine Team

- Led core engine development for a Sim2Real embodied AI platform, spanning robot simulation and large-scale synthetic data generation.
- Built training and deployment infrastructure and a unified computer vision and robotics library for product development.
- Developed physics-based simulation, rendering, and domain-adaptation pipelines for robotic perception and control.

**Pudu Technology Inc.** Dec 2021 - Jul 2022
Computer Vision Engineer

- Developed 6D pose estimation, scene reconstruction, and collision avoidance for catering service robots.
- Automated annotation for synthetic, mixed-reality, and real-world data; researched Sim2Real learning with Tsinghua Shenzhen International Graduate School.

**RVBUST Inc.** Apr 2020 - Dec 2021
Computer Vision Engineer

- Developed point cloud registration, recognition, segmentation, and 6D pose estimation for industrial robot vision.
- Led a cross-functional team developing bin-picking and 3D inspection products.

**Appotronics Inc., Research Institute** Aug 2019 - Apr 2020
Computer Vision Engineer

- Developed projector autofocus and calibration algorithms for camera alignment, screen correction, and distortion compensation.
- Prototyped 3D projection mapping and projected AR systems, and authored related patents.

**Panasonic R&D Centre Singapore** Sep 2018 - Apr 2019
Research Engineer | Internship

- Developed object detection, pose estimation, and point cloud processing for robotic picking.
- Researched deep reinforcement learning for motion planning in simulation. [Code]

## RESEARCH GRANTS & OPEN SOURCE

**Multimodal Databases for Robot Learning of Human Skills** Dec 2024 - Dec 2026
Principal Investigator | RMB 4 million | 深圳市具身智能重大专项
Lead simulator-based data generation, Real2Sim mapping, and multimodal dataset and skill database development for robot learning.

**EmbodiChain**
Project Lead | GitHub | Website
A modular, GPU-accelerated platform for simulation, synthetic data generation, and robot policy training and evaluation, supporting Sim2Real transfer.

**Open3D** Apr 2022 - Present
Open-source Collaborator | GitHub | Website
Collaborate on tensor-based geometry, 3D reconstruction, and the tensor backend of an open-source 3D data processing library.

## PUBLICATIONS AND REPORTS

12. **RiCo: Neural Simulation of Rigid-Body Interactions via Local Contact Reasoning**
Ruixiang Ouyang, Guanren Qiao, Fansen Meng, **Yueci Deng**, Ruixing Jin, Kui Jia, Guiliang Liu†. *arXiv preprint, 2026.*

11. **RoboSynChallenge: Mastering Real-World Dexterity via Generalizing Synthesized Manipulation Skills**
RoboSynChallenge Organizing Team. *NeurIPS Competition Report, 2026.*

10. **PAct: Part-Decomposed Single-View Articulated Object Generation**
Qingming Liu, Xinyue Yao, Shuyuan Zhang, **Yueci Deng**, Guiliang Liu, Zhen Liu†, Kui Jia. *ACM SIGGRAPH Asia, 2026.*

9. **Backpropagating Through Simulation: Analytic Policy Gradients for Sample and Learning Efficient Differentiable Continuous Control**
**Yueci Deng**†. *arXiv preprint, 2026.*

8. **Focus-Then-Contact: Speeding Up Robotic Contact-Rich Task Learning with Affordance-Guided Real-World Residual Reinforcement Learning**
Guanren Qiao, Ruixiang Ouyang, Sheng Xu, Ruixing Jin, **Yueci Deng**, Yunxin Tai, Kui Jia, Guiliang Liu†. *ICML, 2026.*

7. **From Reaction to Anticipation: Proactive Failure Recovery through Agentic Task Graph for Robotic Manipulation**
Sheng Xu, Ruixing Jin, Huayi Zhou, Bo Yue, Guanren Qiao, Yunxin Tai, **Yueci Deng**, Kui Jia, Guiliang Liu†. *RSS, 2026.*

6. **YOTO++: Learning Long-Horizon Closed-Loop Bimanual Manipulation from One-Shot Human Video Demonstrations**
Huayi Zhou, Ruixiang Wang, Yunxin Tai, **Yueci Deng**, Guiliang Liu, Kui Jia†. *IEEE TPAMI, 2026.*

5. **EWA: Aligning Video World Models with Executable Robot Actions via Inverse Dynamics Rewards**
Ruixiang Wang, Qingming Liu, **Yueci Deng**, Guiliang Liu, Zhen Liu†, Kui Jia. *arXiv preprint, 2026.*

4. **Sim2Real VLA: Zero-Shot Generalization of Synthesized Skills to Realistic Manipulation**
Runyi Zhao, Sheng Xu, Ruixing Jin, **Yueci Deng**, Yunxin Tai, Kui Jia, Guiliang Liu†. *ICLR Poster, 2026.*

3. **GS-World: An Engine-driven Learning Paradigm for Pursuing Embodied Intelligence using World Models of Generative Simulation**
Guiliang Liu, **Yueci Deng**, Zhen Liu, Kui Jia†. *Position paper, 2025.*

2. **DexScale: Automating Data Scaling for Sim2Real Generalizable Robot Control**
Guiliang Liu*, **Yueci Deng***, Runyi Zhao, Huayi Zhou, Jian Chen, Jietao Chen, Ruiyan Xu, Yunxin Tai, Kui Jia†. *ICML Poster, 2025.*

1. **You Only Teach Once: Learn One-Shot Bimanual Robotic Manipulation from Video Demonstrations**
Huayi Zhou, Ruixiang Wang, Yunxin Tai, **Yueci Deng**, Guiliang Liu, Kui Jia†. *RSS, 2025.*

* Equal contribution. † Corresponding author.

## TECHNICAL SKILLS

**Programming & tools** C / C++ / CUDA, Python, CMake, Git.
**Research & engineering** 3D vision, robotics, machine learning, computer graphics, differentiable simulation.
**Delivery** Applied R&D, technology productization, large-scale software development.

## PATENTS AND APPLICATIONS

- **CN114155291A** — Box pose recognition method, apparatus, terminal and storage medium.
- **CN114653629B** — Vision-based sorting method, intelligent sorting system and readable storage medium.
- **CN114820781A** — Machine vision-based intelligent handling method, apparatus, system and storage medium.
- **CN217453968U** — Robot, robotic arm and pneumatic gripper.
- **CN121018513A** — Robot URDF model free assembly and constraint management method and system.
- **CN121683799A** — Semantic recognition-based dynamic URDF model construction system and method.
- **CN121934817A** — Robot URDF model construction method, apparatus, electronic device and storage medium.
- **CN122595497A** — Robot model generation method, apparatus, device and medium.
- **CN122574319A** — Grasping pose generation method, apparatus, device and computer-readable storage medium.
- **CN122595647A** — Data processing method, apparatus, computer, storage medium and program product.
- **CN113359381B** — Projection system and projection screen adjustment method.
- **CN113467600B** — Augmented reality-based information display method, system, apparatus and projection device.
- **CN113467601B** — Augmented reality-based information display method, system, apparatus and projection device.
- **CN113497925B** — Projection focusing method and projection focusing device.
- **CN113573035A** — Vision-based adaptive AR-HUD brightness adjustment method.
- **CN113934098B** — Projection device, automatic focusing method and apparatus.
- **CN114286064A** — Real-time focusing method, apparatus, system and computer-readable storage medium.

A: application publication; B: invention grant; U: utility model grant.